Case Study 1: The Ames Goldsmith Decommissioning Disaster

On April 22, 2026, a fatal chemical release occurred at the Ames Goldsmith Catalyst Refiners facility in Institute, West Virginia, during permanent plant decommissioning and dismantling operations. Personnel executing equipment cleaning inside a processing building introduced concentrated nitric acid (HNO_3) and a proprietary surfactant cleaning solution (Bonderite M2000A) into a legacy vessel containing unquantified residues of "silver mud"—a sulphurous sediment consisting predominantly of insoluble silver sulfide (Ag_2S).

The inadvertent mixing triggered a violent acid-base displacement and thermal decomposition reaction that rapidly pressurised the vessel and vented into the enclosed workplace atmosphere. Rather than a standard acid spill, the reaction liberated lethal concentrations of gaseous hydrogen sulfide (H_2S) alongside toxic nitrogen dioxide (NO_2) fumes. Because the facility lacked continuous online toxic gas detection for this transient phase, two operators were rapidly overcome and killed by chemical knockdown, while over thirty others, including seven emergency first responders, were hospitalised with severe inhalational injuries and pulmonary oedema.

Key Questions

1) What went wrong chemically and procedurally during the decontamination sequence?
2) What engineering safeguards and management systems can prevent similar transient mishaps?
3) How should emergency response and containment be managed for non-routine chemical releases?

Deliverables:

- Thermodynamic and Kinetic Reaction Modelling
- HAZOP/LOPA study
- Decommissioning MOC framework
- Continuous Online Gas Detection and Monitoring Protocol
- Etc

Case Study 2: The RINL Visakhapatnam Molten Steel Explosion

On June 8, 2026, a catastrophic industrial disaster occurred inside the Continuous Casting Department (CCD) of Steel Melt Shop-1 (SMS-1) at Rashtriya Ispat Nigam Limited's (RINL) Visakhapatnam Steel Plant in Andhra Pradesh, India. Ladle No. 19, a refractory-lined vessel carrying approximately 150 tonnes of liquid crude steel superheated to 1,500°C–1,600°C, was suspended from an Electric Overhead Travelling (EOT) crane above the casting tundish. As the operator actuated the hydraulic slide-gate mechanism to initiate pouring, the bottom seal ruptured explosively.

The blast was driven by the rapid nucleation and thermal expansion of entrapped purging argon gas and localized "fire bubbles" sealed beneath the immense metallostatic head of liquid metal. The resulting explosion destroyed the sliding gate assembly, showering 150 tonnes of molten steel across an 80-meter radius and igniting the overhead crane infrastructure. Ten steelworkers were killed, six sustained severe third-degree burns, and the facility suffered over INR 120 crore ($14 million USD) in direct structural damage and production halts. Critically, a precursor gas explosion involving a 60-tonne spill had occurred in an adjacent melt shop just 40 minutes prior, but operations were not halted.

Key Questions

1) What physical and operational mechanisms triggered the ladle explosion, and why were warning signals missed?
2) What engineering safeguards and layout modifications will prevent molten steel transfer explosions?

Deliverables

- Root Cause Analysis
- Fluid dynamics / Argon nucleation analysis
- HAZOP/LOPA study
- Warning indicator protocol
- Etc